

PADHAISHURU

GATE Preparation Platform

GATE PE

Petroleum Engineering

TREND-BASED MOCK PAPER

— 04 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	10 subjects

Difficulty Distribution

Easy: 14 questions (22%)

Moderate: 42 questions (65%)

Difficult: 9 questions (14%)

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
These mock papers are constructed using analysis of historical GATE
PYQ patterns.
Historical patterns do not guarantee future examination questions.

GENERAL INSTRUCTIONS

GATE PE — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
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4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1.

1M [no -ve]

[General Aptitude]

15% of 2000:

*Enter your answer here***Q2.**

1M [-0.33]

*[Engineering Mathematics]*First order ODE $dy/dx = y$ solution:

- (A) Ce^x
- (B) Ce^{-x}
- (C) x^2
- (D) $\ln(x)$

Q3.

1M [no -ve]

[Petroleum Exploration]

Source rock rich in:

*Enter your answer here***Q4.**

1M [-0.33]

*[Reservoir Engineering]*Darcy's law: $q = ?$

- (A) $kA\Delta P / 4L$
- (B) $;\ddot{A} \ddot{o}^\circ$
- (C) $kL/;\ddot{A}$
- (D) $;\hat{A}\hat{o}^\circ$

Q5.

1M [-0.33]

[Drilling Engineering]

Roller cone bit has:

- (A) 1 cone
- (B) 2
- (C) 3
- (D) 4

Q6.

1M [-0.33]

[Production Engineering]

ESP stands for:

- (A) Electric Submersible Pump
- (B) Electronic Surface Pump
- (C) Electrostatic Pump
- (D) None

Q7.

1M [-0.33]

[Petroleum Formation Evaluation]

Gamma ray log measures:

- (A) Porosity
- (B) Saturation
- (C) Shale
- (D) Permeability

Q8.

1M [-0.33]

[Well Testing]

Pressure transient analysis uses:

- (A) Horner plot
- (B) Material balance
- (C) MBAL
- (D) PVT

Q9.

1M [-0.33]

[Petroleum Chemistry]

API gravity > 31.1 is:

- (A) Heavy
- (B) Light
- (C) Extra heavy
- (D) Bitumen

Q10.

1M [no -ve]

[Offshore Drilling]

Jack-up rig works in:

*Enter your answer here***Q11.**

1M [-0.33]

[Petroleum Chemistry]

API gravity > 31.1 is:

- (A) Heavy
- (B) Light
- (C) Extra heavy
- (D) Bitumen

Q12.

1M [-0.33]

[Offshore Drilling]

Jack-up rig works in:

- (A) Deep water
- (B) Shallow water
- (C) Both
- (D) None

Q13.

1M [-0.33]

[Offshore Drilling]

Jack-up rig works in:

- (A) Deep water
- (B) Shallow water
- (C) Both
- (D) None

Q14.

1M [-0.33]

[Production Engineering]

ESP stands for:

- (A) Electric Submersible Pump
- (B) Electronic Surface Pump
- (C) Electrostatic Pump
- (D) None

Q15.

1M [-0.33]

[Petroleum Exploration]

Source rock rich in:

- (A) Carbonate
- (B) Organic matter
- (C) Sand
- (D) Shale

Q16.

1M [-0.33]

[General Aptitude]

15% of 2000:

- (A) 200
- (B) 300
- (C) 400
- (D) 150

Q17.

1M [-0.33]

[General Aptitude]

15% of 2000:

- (A) 200
- (B) 300
- (C) 400
- (D) 150

Q18.

1M [-0.33]

*[Reservoir Engineering]*Darcy's law: $q = ?$

- (A) $kA \frac{\Delta P}{\mu L}$
- (B) $\frac{kA \Delta P}{\mu L}$
- (C) $\frac{kL}{\mu} \Delta P$
- (D) $\frac{kA \Delta P}{\mu}$

Q19.

1M [-0.33]

[General Aptitude]

15% of 2000:

- (A) 200
- (B) 300
- (C) 400
- (D) 150

Q20.

1M [-0.33]

[Offshore Drilling]

Jack-up rig works in:

- (A) Deep water
- (B) Shallow water
- (C) Both
- (D) None

Q21.

1M [-0.33]

[General Aptitude]

15% of 2000:

- (A) 200
- (B) 300
- (C) 400
- (D) 150

Q22.

1M [-0.33]

[General Aptitude]

15% of 2000:

- (A) 200
- (B) 300
- (C) 400
- (D) 150

Q23.

1M [-0.33]

[General Aptitude]

15% of 2000:

- (A) 200
- (B) 300
- (C) 400
- (D) 150

Q24.

1M [-0.33]

[Production Engineering]

ESP stands for:

- (A) Electric Submersible Pump
- (B) Electronic Surface Pump
- (C) Electrostatic Pump
- (D) None

Q25.

1M [-0.33]

[Offshore Drilling]

Jack-up rig works in:

- (A) Deep water
- (B) Shallow water
- (C) Both
- (D) None

Q26.

1M [-0.33]

*[Engineering Mathematics]*First order ODE $dy/dx = y$ solution:

- (A) Ce^x
- (B) Ce^{-x}
- (C) x^2
- (D) $\ln(x)$

Q27.

1M [-0.33]

[Petroleum Chemistry]

API gravity > 31.1 is:

- (A) Heavy
- (B) Light
- (C) Extra heavy
- (D) Bitumen

Q28.

1M [-0.33]

[Well Testing]

Pressure transient analysis uses:

- (A) Horner plot
- (B) Material balance
- (C) MBAL
- (D) PVT

Q29.

1M [-0.33]

*[Reservoir Engineering]*Darcy's law: $q = ?$

- (A) $kA \frac{\Delta P}{L}$
- (B) $\frac{\Delta P}{kA L}$
- (C) $kL \Delta P$
- (D) $\frac{\Delta P}{kA}$

Q30.

1M [no -ve]

[Petroleum Chemistry]

API gravity > 31.1 is:

Enter your answer here

Q31.

2M [no -ve]

*[Engineering Mathematics]*First order ODE $dy/dx = y$ solution:

*Enter your answer here***Q32.**

2M [-0.66]

[Petroleum Exploration]

Source rock rich in:

- (A) Carbonate
- (B) Organic matter
- (C) Sand
- (D) Shale

Q33.

2M [-0.66]

[Petroleum Exploration]

Source rock rich in:

- (A) Carbonate
- (B) Organic matter
- (C) Sand
- (D) Shale

Q34.

2M [no -ve]

[Offshore Drilling]

Jack-up rig works in:

*Enter your answer here***Q35.**

2M [-0.66]

[General Aptitude]

15% of 2000:

- (A) 200
- (B) 300
- (C) 400
- (D) 150

Q36.

2M [-0.66]

[Production Engineering]

ESP stands for:

- (A) Electric Submersible Pump
- (B) Electronic Surface Pump
- (C) Electrostatic Pump
- (D) None

Q37.

2M [no -ve]

[Offshore Drilling]

Jack-up rig works in:

Enter your answer here

Q38.

2M [-0.66]

[Production Engineering]

ESP stands for:

- (A) Electric Submersible Pump
- (B) Electronic Surface Pump
- (C) Electrostatic Pump
- (D) None

Q39.

2M [no -ve]

[General Aptitude]

15% of 2000:

*Enter your answer here***Q40.**

2M [-0.66]

[Petroleum Formation Evaluation]

Gamma ray log measures:

- (A) Porosity
- (B) Saturation
- (C) Shale
- (D) Permeability

Q41.

2M [-0.66]

*[Engineering Mathematics]*First order ODE $dy/dx = y$ solution:

- (A) Ce^x
- (B) Ce^{-x}
- (C) x^2
- (D) $\ln(x)$

Q42.

2M [no -ve]

[Petroleum Exploration]

Source rock rich in:

*Enter your answer here***Q43.**

2M [no -ve]

[Petroleum Exploration]

Source rock rich in:

*Enter your answer here***Q44.**

2M [-0.66]

[Well Testing]

Pressure transient analysis uses:

- (A) Horner plot
- (B) Material balance
- (C) MBAL
- (D) PVT

Q45.

2M [no -ve]

[Offshore Drilling]

Jack-up rig works in:

- (A) Deep water
- (B) Shallow water
- (C) Both
- (D) None

Q46.

2M [-0.66]

[Well Testing]

Pressure transient analysis uses:

- (A) Horner plot
- (B) Material balance
- (C) MBAL
- (D) PVT

Q47.

2M [-0.66]

[General Aptitude]

15% of 2000:

- (A) 200
- (B) 300
- (C) 400
- (D) 150

Q48.

2M [-0.66]

*[Reservoir Engineering]*Darcy's law: $q = ?$

- (A) $kA \frac{\Delta p}{L}$
- (B) $\frac{\Delta p}{kA L}$
- (C) $kL / \Delta p$
- (D) $\frac{\Delta p}{kL}$

Q49.

2M [-0.66]

*[Reservoir Engineering]*Darcy's law: $q = ?$

- (A) $kA \frac{\Delta p}{L}$
- (B) $\frac{\Delta p}{kA L}$
- (C) $kL / \Delta p$
- (D) $\frac{\Delta p}{kL}$

Q50.

2M [-0.66]

[Offshore Drilling]

Jack-up rig works in:

- (A) Deep water
- (B) Shallow water
- (C) Both
- (D) None

Q51.

2M [-0.66]

[Petroleum Chemistry]

API gravity > 31.1 is:

- (A) Heavy
- (B) Light
- (C) Extra heavy
- (D) Bitumen

Q52.

2M [-0.66]

[General Aptitude]

15% of 2000:

- (A) 200
- (B) 300
- (C) 400
- (D) 150

Q53.

2M [-0.66]

[Offshore Drilling]

Jack-up rig works in:

- (A) Deep water
- (B) Shallow water
- (C) Both
- (D) None

Q54.

2M [-0.66]

*[Engineering Mathematics]*First order ODE $dy/dx = y$ solution:

- (A) Ce^x
- (B) Ce^{-x}
- (C) x^2
- (D) $\ln(x)$

Q55.

2M [no -ve]

[General Aptitude]

15% of 2000:

*Enter your answer here***Q56.**

2M [no -ve]

*[Engineering Mathematics]*First order ODE $dy/dx = y$ solution:

*Enter your answer here***Q57.**

2M [-0.66]

[Well Testing]

Pressure transient analysis uses:

- (A) Horner plot
- (B) Material balance
- (C) MBAL
- (D) PVT

Q58.

2M [-0.66]

[Petroleum Exploration]

Source rock rich in:

- (A) Carbonate
- (B) Organic matter
- (C) Sand
- (D) Shale

Q59.

2M [no -ve]

[Petroleum Formation Evaluation]

Gamma ray log measures:

*Enter your answer here***Q60.**

2M [no -ve]

[Petroleum Formation Evaluation]

Gamma ray log measures:

*Enter your answer here***Q61.**

2M [no -ve]

*[Engineering Mathematics]*First order ODE $dy/dx = y$ solution:

*Enter your answer here***Q62.**

2M [no -ve]

*[Reservoir Engineering]*Darcy's law: $q = ?$

*Enter your answer here***Q63.**

2M [-0.66]

[Drilling Engineering]

Roller cone bit has:

- (A) 1 cone
- (B) 2
- (C) 3
- (D) 4

Q64.

2M [-0.66]

[Petroleum Exploration]

Source rock rich in:

- (A) Carbonate
- (B) Organic matter
- (C) Sand
- (D) Shale

Q65.

2M [no -ve]

[Reservoir Engineering]

Darcy's law: $q = ?$

Enter your answer here